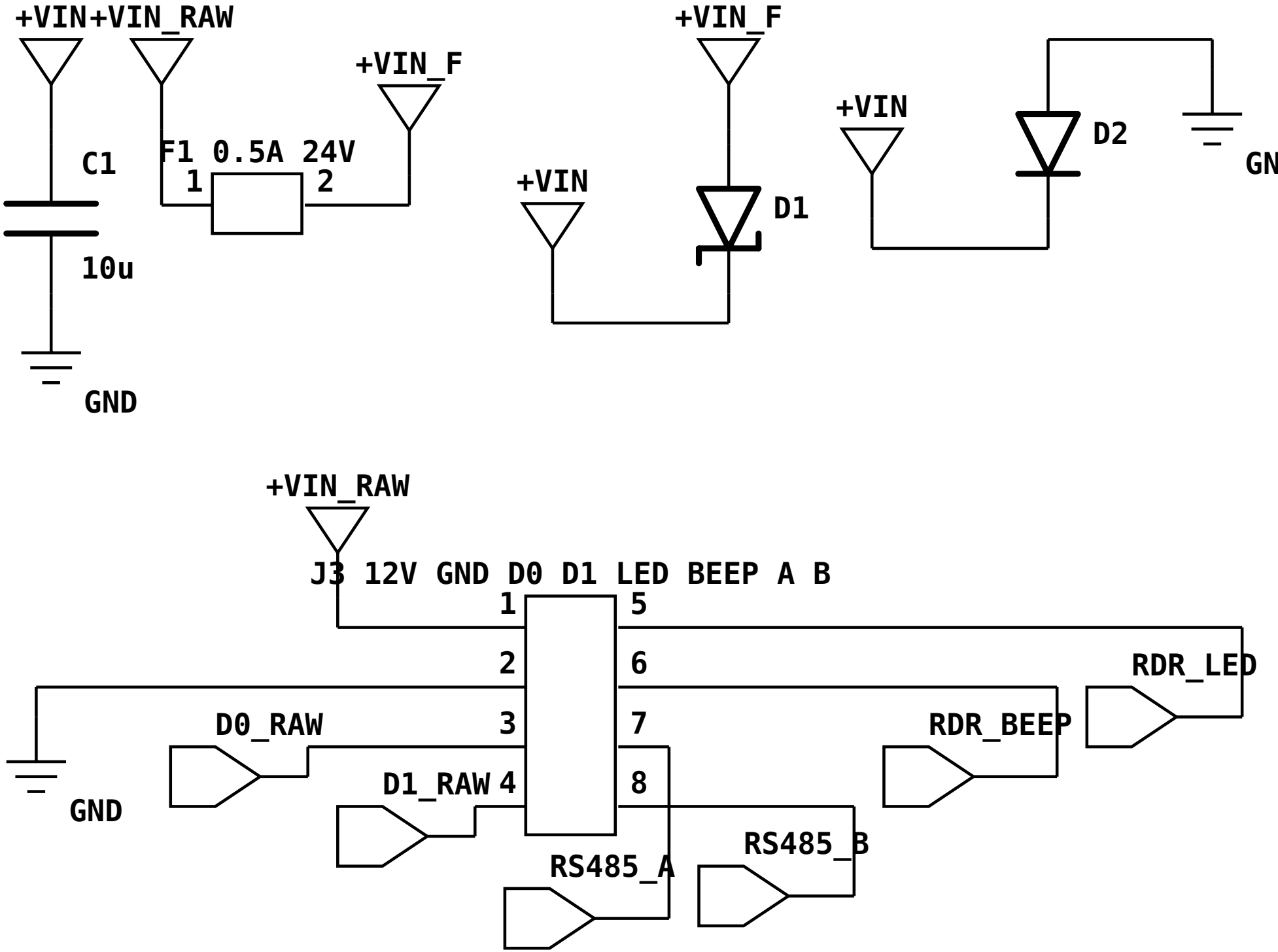
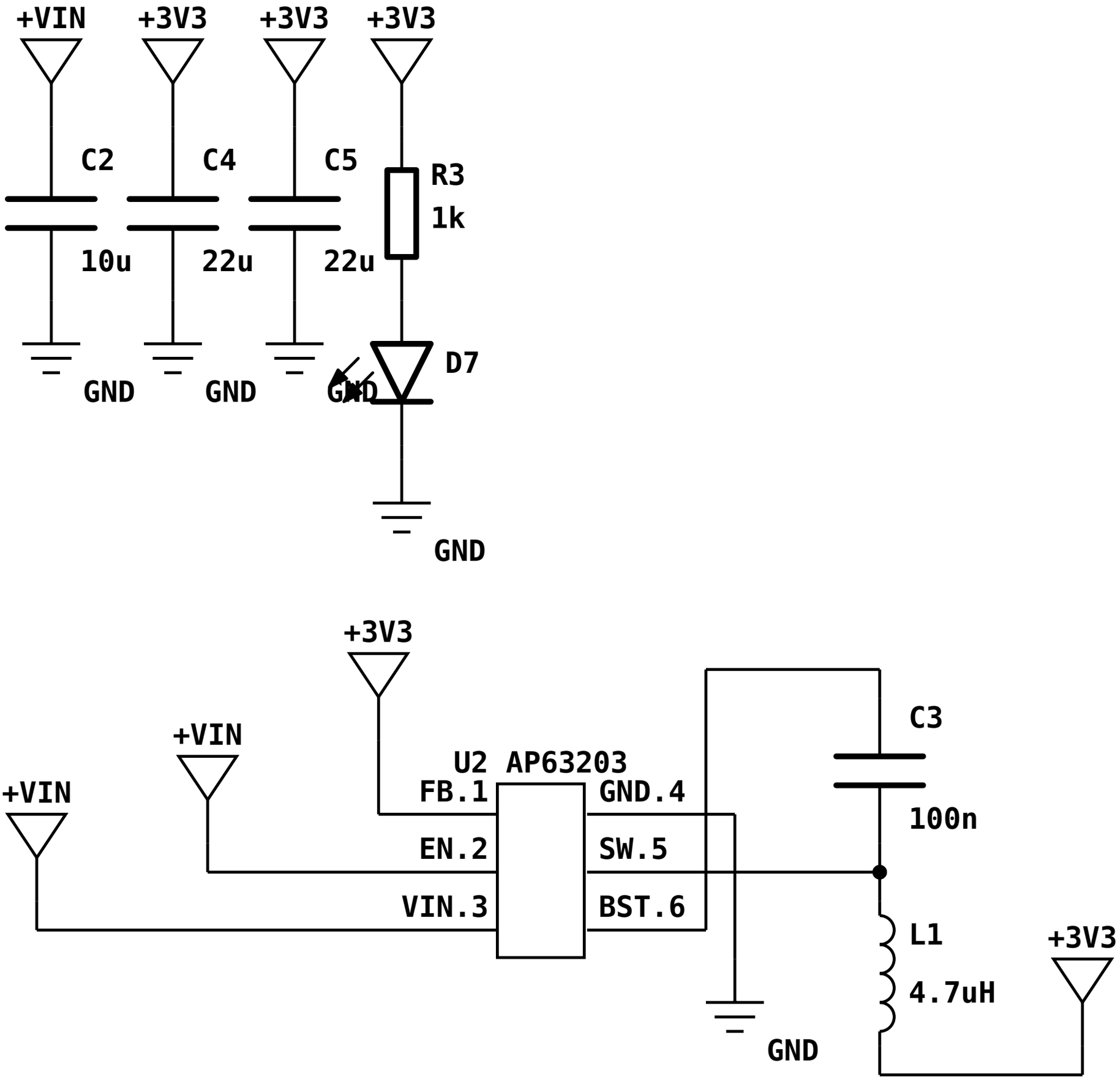
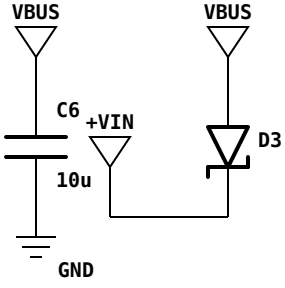
Wiegand bridge LITE — revision 1 schematic

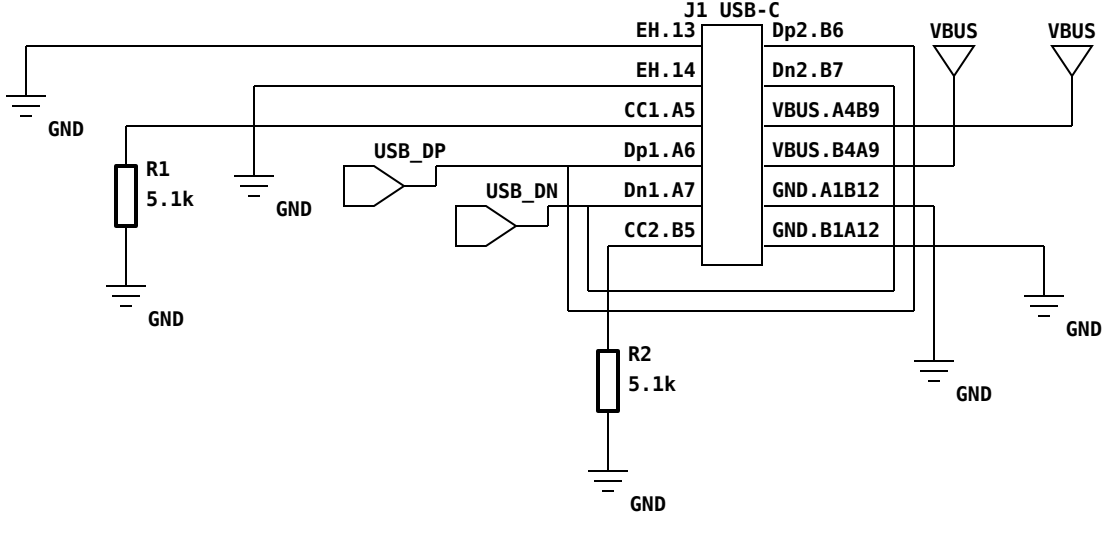
Generated from netlist.py (netlistsvg). Supply nets are flags; nets shared between blocks are named ports. I

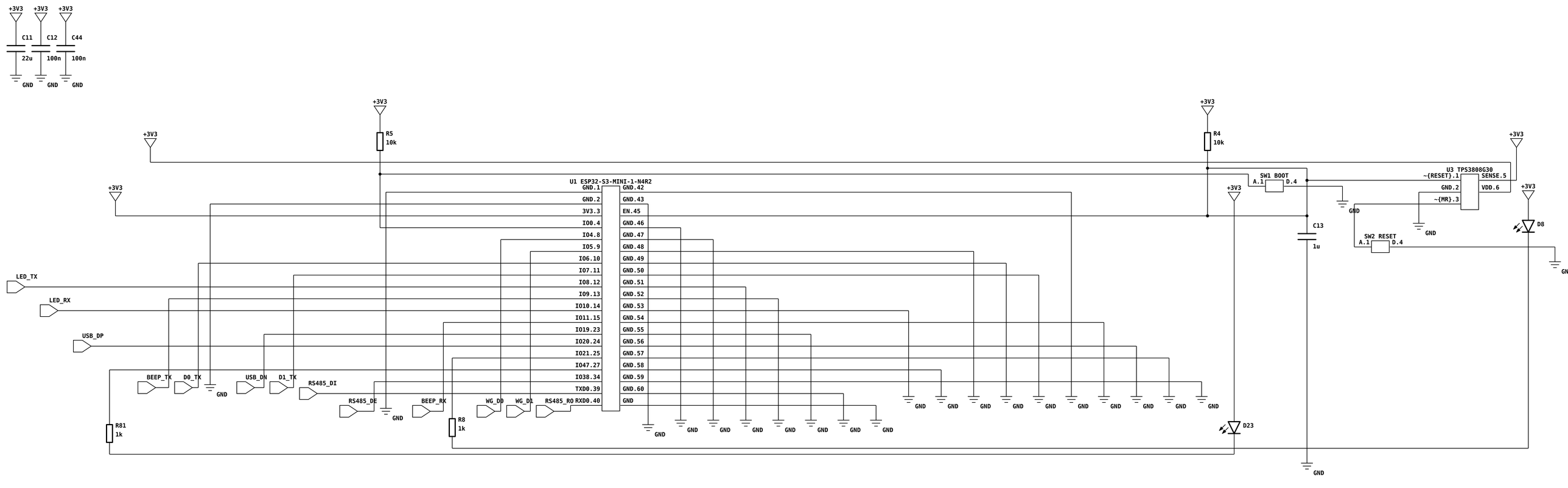
1. Block 1: terminal block and 8-16 V input
2. Block 2: buck to 3.3 V (AP63203, datasheet figure 21)
3. Block 3: USB-C (programming, and bench power through D3)
4. Block 4: MCU, buttons, status LEDs
5. Block 5: reader lines D0, D1, LED, BEEP -- bidirectional cells
6. Block 6: RS-485 / OSDP on UART0 (rev 10 block 14, less the bias network)
7. Block 7: surge/ESD on every field wire (SM712: +12 / -7 V standoff)

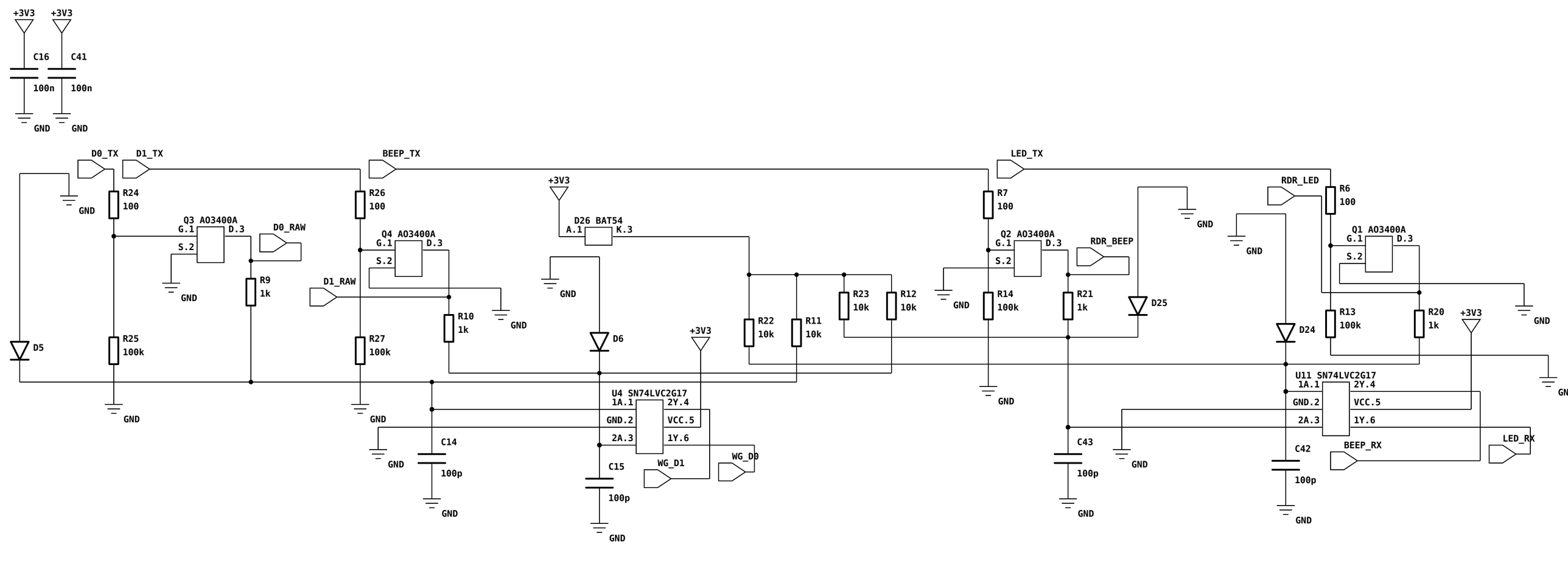

+VIN
+VIN_RAW
+VIN_F
C1
F1 0.5A 24V
1
2
10u
GND
+VIN_F
+VIN
D1
+VIN
D2
GN
+VIN_RAW
J3 12V GND D0 D1 LED BEEP A B
1
2
3
4
5
6
7
8
RDR_LED
D0_RAW
RDR_BEEP
GND
D1_RAW
RS485_B
RS485_A

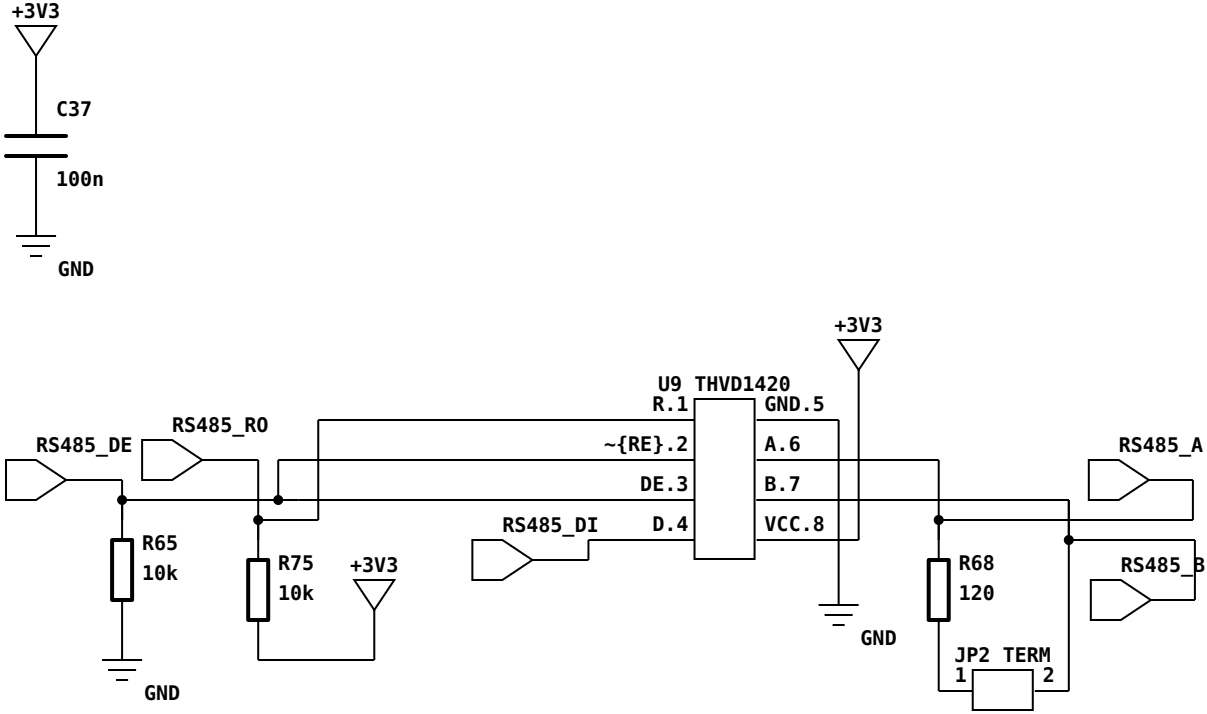

+VIN
+3V3
+3V3
+3V3
C2
C4
C5
R3
1k
10u
22u
22u
D7
GND
GND
GND
GND
+3V3
+VIN
+VIN
U2 AP63203
FB.1
EN.2
VIN.3
GND.4
SW.5
BST.6
C3
100n
L1
4.7uH
+3V3
GND

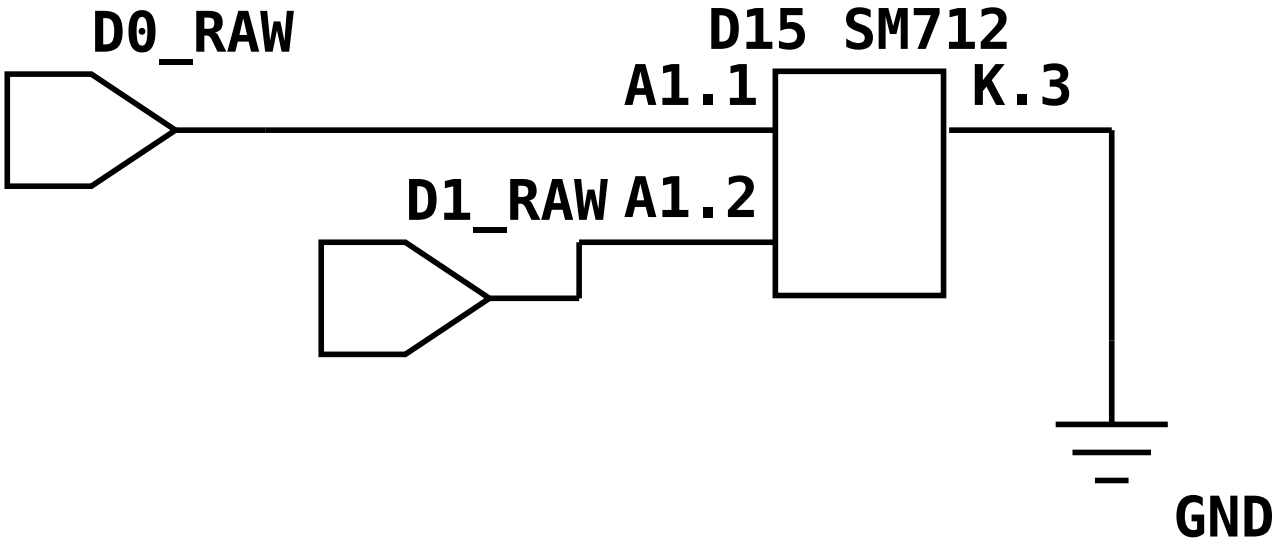

VBUS
VBUS
C6
+VIN
D3
10u
GND

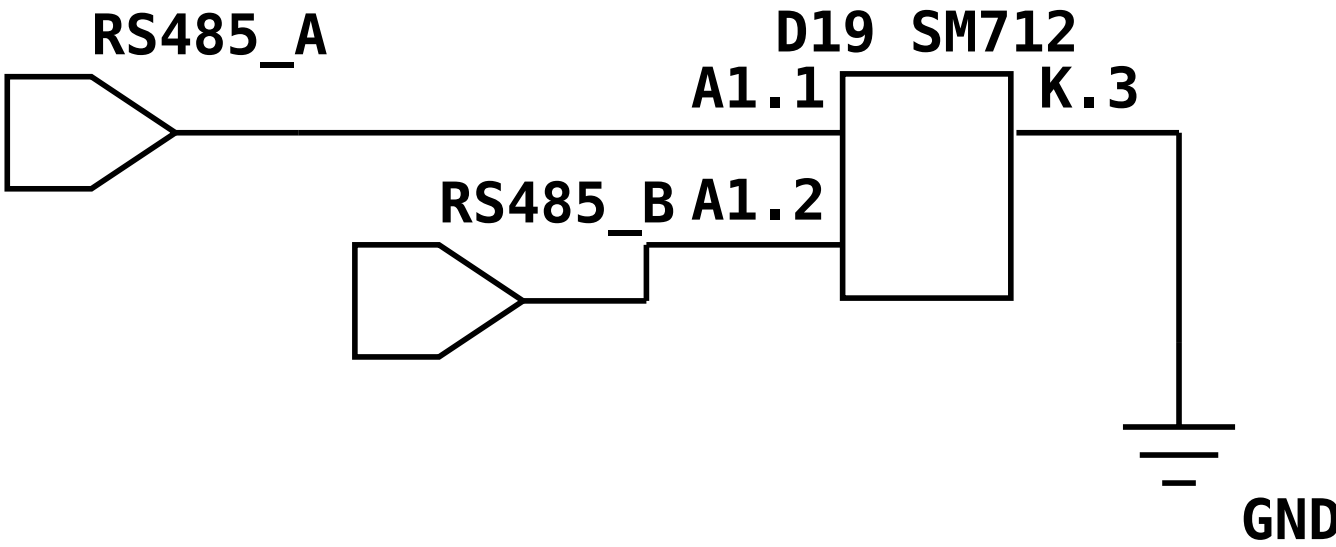

J1 USB-C
EH.13
Dp2.B6
EH.14
Dn2.B7
CC1.A5
VBUS.A4B9
Dp1.A6
VBUS.B4A9
Dn1.A7
GND.A1B12
CC2.B5
GND.B1A12
VBUS
VBUS
GND
R1
5.1k
GND
USB_DP
USB_DN
GND
GND
R2
5.1k
GND

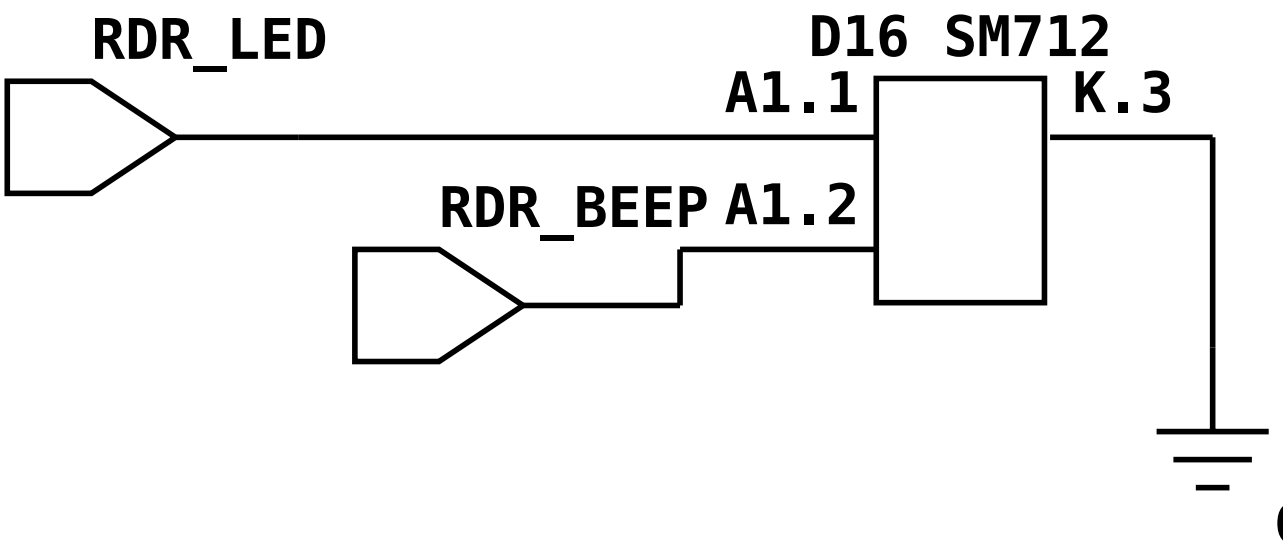

+3V3
C11
22u
GND
C12
100n
C44
100n
R5
10k
R4
10k
U1 ESP32-S3-MINI-1-N4R2
GND.1
GND.2
3V3.3
IO0.4
IO4.8
IO5.9
IO6.10
IO7.11
IO8.12
IO9.13
IO10.14
IO11.15
IO19.23
IO20.24
IO21.25
IO47.27
IO38.34
TXD0.39
GND
GND.42
GND.43
EN.45
GND.46
GND.47
GND.48
GND.49
GND.50
GND.51
GND.52
GND.53
GND.54
GND.55
GND.56
GND.57
GND.58
GND.59
GND.60
RXD0.40
SW1 BOOT
A.1
D.4
U3 TPS3808G30
~{RESET}.1
SENSE.5
GND.2
VDD.6
~{MR}.3
C13
1u
SW2 RESET
D8
LED_TX
LED_RX
USB_DP
BEEP_TX
D0_TX
USB_DN
D1_TX
RS485_DI
RS485_DE
RS485_RO
BEEP_RX
WG_D0
WG_D1
R81
1k
R8
1k
D23


+3V3
+3V3
C16
C41
100n
100n
GND
GND
D0_TX
D1_TX
BEEP_TX
LED_TX
RDR_LED
R24
100
R26
100
R7
100
R6
100
Q3 AO3400A
G.1
D.3
S.2
D0_RAW
Q4 AO3400A
D26 BAT54
A.1
K.3
Q2 AO3400A
RDR_BEEP
Q1 AO3400A
R9
1k
D1_RAW
R10
1k
R23
10k
R12
10k
R14
100k
R21
1k
D25
R13
100k
R20
1k
D5
R25
100k
R27
100k
D6
R22
10k
R11
10k
D24
U4 SN74LVC2G17
1A.1
2Y.4
GND.2
VCC.5
2A.3
1Y.6
U11 SN74LVC2G17
C14
100p
C15
100p
C43
100p
C42
100p
WG_D1
WG_D0
BEEP_RX
LED_RX


+3V3
C37
100n
GND
+3V3
U9 THVD1420
R.1
GND.5
RS485_RO
~{RE}.2
A.6
RS485_A
RS485_DE
DE.3
B.7
RS485_DI
D.4
VCC.8
R65
10k
R75
10k
+3V3
R68
120
RS485_B
GND
GND
JP2 TERM
1
2


D0_RAW
D15 SM712
A1.1
K.3
D1_RAW
A1.2
GND


RS485_A
D19 SM712
A1.1
K.3
RS485_B
A1.2
GND


RDR_LED
D16 SM712
A1.1
K.3
RDR_BEEP
A1.2
GN